

WATER GUARDIAN

Pre-Purchase Technical Feasibility & Simulation Report

Version 1.0 · 2026-09-12 · Workspace `water-guardian-simulation/`

Tags used throughout: **VERIFIED** (primary document read, or listing seen) · **CALCULATED** (pure arithmetic/geometry) · **SIMULATED** (proven in the digital test bench or host tests) · **ASSUMED** (our engineering assumption) · **NOT YET PHYSICALLY VERIFIED** (only a bench test with a real meter can settle it). Research-access caveat: manufacturer, marketplace and government pages were mostly blocked from the research sandbox; facts from those pages are quoted via search snippets and flagged (*snippet*). The TUF-2000M manual itself was read in full from a GitHub mirror.

1. Executive Summary

Can this system actually work? The electronics, protocol, firmware and detection logic: **yes, with evidence**. The one thing that decides the project — whether a ~SAR 600 clamp-on transit-time meter reads a small PPR pipe at 0.5–1 L/min with a stable zero — **is not yet verified, and the available field evidence is unfavourable**.

- **VERIFIED:** the TUF-2000M has an isolated RS485 Modbus RTU port, runs on 8–36 VDC at 50 mA, publishes a complete register map with flow (REAL4), signal quality, strengths, transit ratio and error bits, and a working-step letter that tells us when it has no signal. Three open-source projects read it from ESP32/Python. Default protocol is Modbus **ASCII** (must set M63 = RTU). PPR is **not** in its material list.
- **CALCULATED:** on a DN25 PPR PN20 pipe (ID ≈ 16.6 mm) the meter's default 0.03 m/s low-flow cutoff equals $\approx$ **0.39 L/min**; on a 2" riser it equals $\approx$ **3.9 L/min**. Sensor placement on a small branch is therefore mandatory; the main riser is blind to small flows.
- **SIMULATED:** a 25-scenario campaign passes every logic scenario (sensor failure $\rightarrow$ SENSOR_UNKNOWN never zero; offline detection with later sync; reboot restore; cleaning override; occupied-period REVIEW; resolution) under three profiles; firmware host tests pass 37/37. The 0.25 L/min rung is **undetectable** under the published-spec profile, by geometry. Under a **pessimistic profile derived from user reports** (zero offset +0.03 m/s, noise 0.05 m/s), an *uncalibrated* meter produces **false incidents and incidents that never resolve**; a zero calibration (M42+M26) plus 30 s averaging removes them at a 0.5 L/min threshold, and a 1.0 L/min threshold removes them even without calibration.
- **NOT YET PHYSICALLY VERIFIED:** TS-2 signal quality on PPR wall (soft, attenuating, not in the material list); real zero-flow noise on DN25; minimum detectable flow; long-term couplant stability.

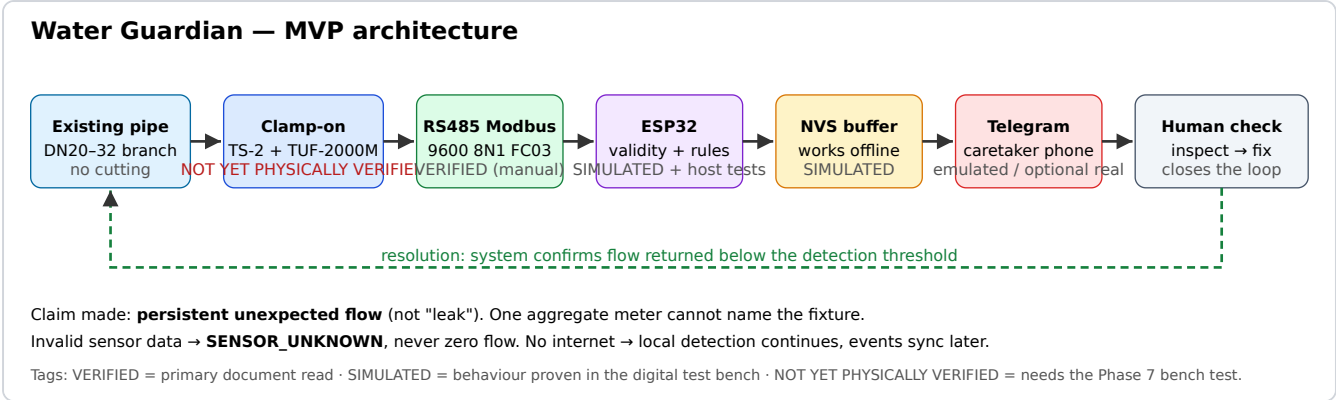
Decision: GO WITH CONDITIONS (§24) — and for the purchase itself: **NO — BORROW FIRST; if a loan is impossible within two weeks, YES — BUT ONLY AFTER** the pipe facts in §23 are in hand and the order is placed as the AliExpress SAR $\approx$ 450–600 kit, budgeted as a *test article* whose failure is an acceptable outcome (§25).

2. Project Objective

Retrofit monitoring of an existing facility pipe (mosque ablution supply first) to detect **persistent unexpected water flow** — not "leak" — without cutting or drilling, on the smallest, cheapest, fastest

credible prototype, using the ESP32 already owned. Reduce technical uncertainty before spending ~SAR 600+ on a clamp-on ultrasonic meter.

3. Final MVP Architecture



```
Qualified existing pipe (DN20-32 branch, full, ≥100/50 straight)
├─ acoustic, no cutting
├─ TS-2 clamp-on transducers, V-method, couplant, side of pipe
├─ 1 coax
├─ TUF-2000M module (8-36 VDC → M63=RTU → M46=I) · 9600 8N1
├─ RS485 A/B (isolated port)
├─ RS485+IIL 3.3 V (MAX3485 with DE/RE, or DFR0845 isolated auto-direction)
├─ 1 UART2 GPIO16/17
├─ ESP32 (owned): Modbus poll 1 s → validity policy (step, 0, strength, ratio, error bits)
├─ 1 valid samples only ── 1 invalid → SENSOR_UNKNOWN (never zero)
├─ 30 s moving average → rule engine (QUIET) / OCCUPIED / CLEANING, threshold + persistence
├─
├─ NVS: incident + unsynced events (survive reboot, work offline)
├─ 1 Wi-Fi when available
├─ Telegram bot → caretaker phone (+ optional HTTP backend/dashboard)
├─
├─ Human inspection → flow stops → system confirms flow < threshold for resolveMin → INCIDENT
├─ RESOLVED
```

Changes vs. the initial proposal, and why: (1) an explicit **validity layer** before the rule engine, because the meter reports 0 when it has no signal and, by default, *holds the last good value* on poor signal (M28) — both would be silently misread as facts; (2) a **30 s moving average and gap tolerance**, because the first simulation run showed noise spikes being counted as "persistence"; (3) **placement on a branch, not the riser**, from the hydraulic table. Diagrams: `diagrams/architecture.svg` , `diagrams/wiring.svg` , `diagrams/architecture.md` (Mermaid state machine).

4. Competition Requirements

(All snippet-level; official pages must be opened to confirm — see `research/notes/competition.md`.)

Item	MEWA University Innovation Challenge (تحدي الابتكار الجامعي)	Future Makers: AI Challenge for Critical Sustainability	SWA Global Prize for Innovation in Water
Organizer	MEWA Agency for Research & Innovation — VERIFIED (snippet)	DCO + MEWA + RDIA + KACST via NexaBridge — REPORTED (press)	Saudi Water Authority — VERIFIED (snippet)
Fit	Best match: water element explicitly "water loss and grey water management"	Only if a genuine AI component exists (AI is mandatory)	Conservation / digital water management categories; global, professional
Team	"1 faculty member and 4-5 students" — VERIFIED (snippet)	innovators/startups/researchers in DCO states; no team rule	universities eligible; no team rule
Edition / deadline	2nd edition reported (Sabq); page updated 2025-07-20; 2026 cycle status and deadline NOT FOUND	closes 4 Oct 2026 (press)	8 Jun–30 Sep 2026; TRL 1–3 / 4–7 tracks
AI required	NOT FOUND → treat as not required	Yes	No
Prototype required	NOT YET VERIFIED ("early-stage ideas" → "pilot projects")	implied (demonstration/testing in KSA)	by track
Evaluation criteria	NOT FOUND	NOT FOUND	NOT FOUND
Restrictions relevant to us	none found; utility-vs-end-user scope of "water loss" unverified	pure threshold logic likely insufficient	none found

Three programmes share the words "تحدي الابتكار" (MEWA UIC, MEWA "للاستدامة" with Babson, MoE "للتنمية المستدامة"); rules must not be mixed. **Action:** open the MEWA UIC page (EN/AR) and the SEU notice from an unblocked browser and record platform URL, dates and criteria before the next milestone.

5. Existing Technology / Market Reality

Clamp-on transit-time meters are a mature industrial product (Flexim, Siemens, KROHNE, E+H) priced SAR 20k–60k, and a Chinese OEM family (TUF-2000 / TDS-100, Dalian Taijia and clones) priced SAR 400–2,000 that shares one firmware lineage, menu map and Modbus table. Small-pipe ($\leq \text{DN}32$) clamp-on is the hard corner of the market: industrial vendors quote $\geq 0.3 \text{ m/s}$ for full accuracy; Keyence sells purpose-built dry-clamp sensors for $\varnothing 13\text{--}44 \text{ mm}$ at SAR 2–4k; hobbyist reports of the cheap family on 22–28 mm pipe describe zero-flow oscillation and drift. There is no evidence anyone sells a sub-SAR-1,000 clamp-on meter *specified* for PPR. Detailed candidates: `SENSOR_COMPARISON.md`, `research/notes/alternatives_and_procurement.md`.

6. Sensor Research

The 33-item questionnaire for the TUF-2000M family, answered from the manual (v13.44) and cross-checked code, is in [research/notes/tuf2000m.md](#) . Headlines:

Item	Finding	Tag
Transducer for DN20–32	TS-2 "small" — DN15–100 (most listings) vs DN25–100 (older listings); manual lists TS-1/TM-1/TL-1 codes, not TS-2	CONFLICT (snippet)
Materials (M14)	carbon steel, stainless, cast/ductile iron, copper, PVC, aluminium, asbestos, fiberglass, other (enter sound speed) . PPR, CPVC, galvanized not listed (galvanized → carbon steel; PPR → "other" + sound speed, value not in any TUF document)	VERIFIED (manual)
Accuracy / repeatability	"better than 1 %" / "better than 0.2 %"; linearity 0.5 % (datasheets)	VERIFIED / snippet
Velocity range	±0.01–12 m/s (vendor); M41 low-flow cutoff default 0.03 m/s (manual); "0.001 m/s with PI transducers" (manual, not clamp-on)	CONFLICT
Straight pipe	upstream 10D (up to 20–30D after fittings/pump), downstream 5D; pipe must be full	VERIFIED
Temperature	TS-2 –30...90 °C (HT –30...160 °C)	snippet
Power	8–36 VDC @ 50 mA (or 10–30 VAC); mains must never be applied to the module	VERIFIED
Interfaces	isolated RS485 (Modbus RTU/ASCII, FC03/06), 4–20 mA loop, 2× OCT (pulse/frequency/relay-like), PT100 inputs	VERIFIED
Pulse	OCT totalizer pulse, unit × multiplier, ≤ 1 pulse per 500 ms period, 6–1000 ms width	VERIFIED
Diagnostics	reg 92 = step (hi byte) + Q 0–99 (lo byte); reg 93/94 strength 0–2047; reg 97–98 transit ratio (normal 100 ± 3 %); reg 72 error bits (bit0 no signal ... bit3 empty pipe); step letters R/I/H/J/G/K/E/Q/F	VERIFIED
Update rate	measurement period 500 ms ; M40 damping default 10 s; first 60 s after power-on is settling	VERIFIED
Low-flow cutoff / hold	M41 (velocity, default 0.03 m/s); M28 "hold last good value" default YES ; M.5 Q-threshold zeroes flow on airy pipes	VERIFIED
Zero	M42 with still water; lost at power-cycle unless M26 saves it	VERIFIED
Couplant / mounting	"adequate coupler... no gap", sand the pipe, transducers on the side to avoid bubbles, V-method 15–200 mm, W 15–50 mm	VERIFIED
Air sensitivity	empty-pipe code K, "poor signal" H; bubbles degrade Q	VERIFIED (qualitative)

Item	Finding	Tag
Price / availability (KSA)	AliExpress TS-2 kit \$100.70–208.50; landed SAR $\approx$ 430–1,090 incl. 15 % VAT; amazon.sa has only handheld TUF-2000H (SAR 502.50); noon: none	snippet / ESTIMATE
Returns	amazon.sa 15 days; AliExpress buyer protection (non-delivery); return-to-China impractical	snippet
Small-pipe field reports	28 mm copper: "huge oscillations around zero when there is no flow", drift, "not suitable for relatively small copper pipes"; 22 mm: hard to get good Q; libe.net: needs Q > 85, read 1.3× low until M45 corrected; 1" PVC: worked. No PPR report found.	snippet (SECONDARY)

7. Selected Sensor Candidates

Scored table in `SENSOR_COMPARISON.md`. Picks: **Best overall / best budget:** TUF-2000M + TS-2. **Best if borrowed:** Flexim FLUXUS F401 or Siemens FS230 portable (distributor demo), or the KFUPM GUNT HM 500 bench as a calibration reference. **Best documented:** Keyence FD-Q (2× budget, no Modbus RTU). **Avoid:** inline sensors as the primary (YF-S201), tubing-only clamp-ons (Sonotec CO.55), > SAR 25k instruments.

8. Ultrasonic Measurement Limitations

(*VERIFIED from the manual and physics of transit-time meters; magnitudes on our pipe are NOT YET PHYSICALLY VERIFIED.*)

- Transit-time difference is tiny at low velocity.** The meter measures Δt between up- and downstream pulses; at 0.03 m/s across a 16 mm bore Δt is in the tens of picoseconds range, so zero stability and electronics noise, not accuracy %, dominate at our flows.
- Wall entry is the weak point on plastics.** PPR has low acoustic impedance and high attenuation relative to metal; a PN20 wall is 3.4–4.2 mm — thick for its diameter. The manual has no PPR entry. Expect lower strength/Q than PVC. Vendor listings' "PVC/plastic" is not a PPR test.
- Small diameters give short acoustic paths** (V-method on 16–21 mm bore), which is why users needed unusual transducer spacings on $\frac{3}{4}$ " copper.
- Zero offset is installation-specific and drifts** with temperature and coupling; the manual requires zeroing on still water and saving with M26.
- The meter is opaque about confidence unless you read Q, strength, ratio and error bits** — which the firmware does; and it can *freeze the last value* (M28) — which the firmware detects only via those diagnostics.
- Air, partial fill and vibration** corrupt readings; the pipe must be full and transducers on the side.
- A single aggregate sensor cannot identify the fixture;** the product claim must stay "persistent unexpected flow".

9. Pipe Compatibility

Material	Meter list (M14)	Our confidence	Note
PVC / uPVC	listed	medium-high	1" Sch40 PVC field report worked

Material	Meter list (M14)	Our confidence	Note
CPVC	not listed (use PVC or "other")	medium	thin CTS wall helps
Copper	listed	medium	but this is where the small-pipe noise reports come from
Steel	listed	medium-high	best acoustic match
Galvanized steel	not listed (use carbon steel)	medium	zinc layer/scale, thick wall; older mosque risers
PPR	not listed — M14 = "other" + sound speed in M15 (value NOT FOUND in TUF docs)	low until tested	the pilot's most likely pipe
Unknown	—	low	measure and identify first

Required before purchase (§23): pipe **material, OD, wall/PN class, ID, location of a full, straight, accessible run, and whether it is the riser or the ablution branch.**

10. Hydraulic Calculations

CALCULATED ($Q = v \cdot A$, $A = \pi D^2/4$; `tests/results/HYDRAULICS_TABLE.md`, `tests/hydraulics_table.js`):

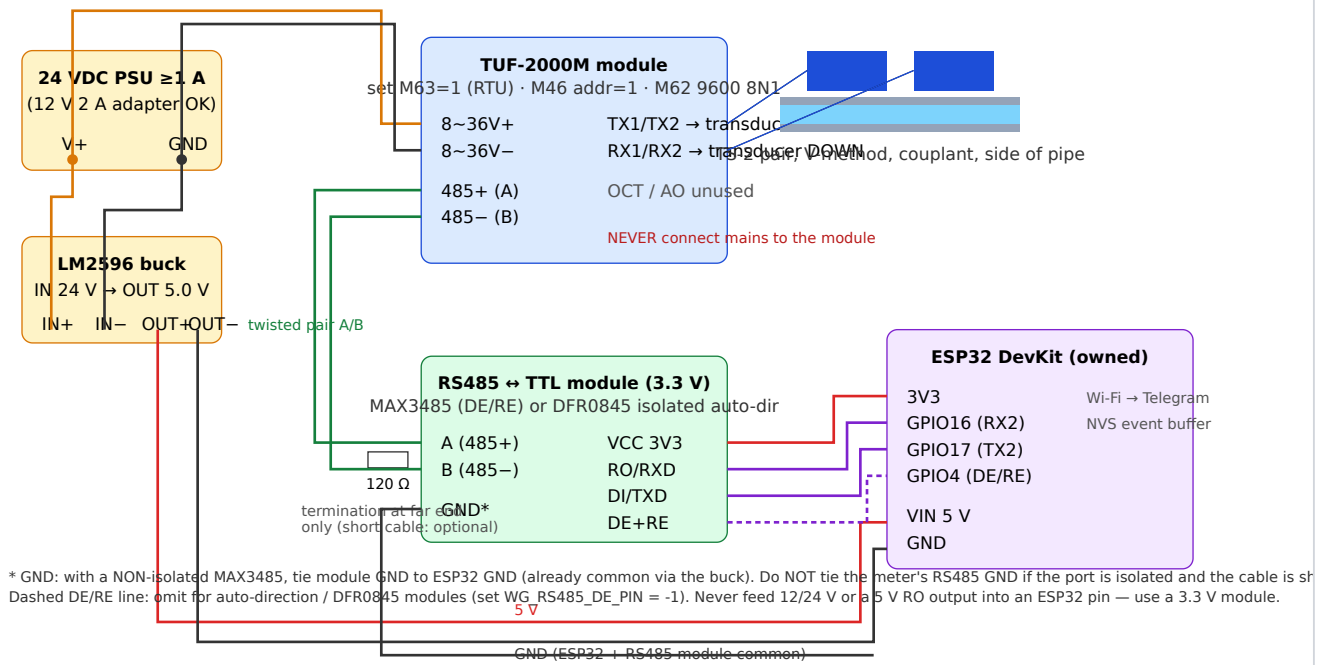
Flow (L/min)	ID 13.2 (PPR 20)	ID 16.6 (PPR 25)	ID 21.2 (PPR 32)	ID 26.6 (GI 1")	ID 52.5 (GI 2")
0.25	0.030 m/s	0.019	0.012	0.007	0.002
0.5	0.061	0.039	0.024	0.015	0.004
1	0.122	0.077	0.047	0.030	0.008
2	0.244	0.154	0.094	0.060	0.015
5	0.609	0.385	0.236	0.150	0.038

Minimum flow at the default **0.03 m/s** cutoff: PPR 20 → 0.25 L/min · PPR 25 → **0.39** · PPR 32 → 0.64 · GI 1" → 1.0 · GI 2" → **3.9 L/min**. Same volumetric flow, 10× lower velocity on the riser — the riser cannot see a dripping tap; a DN20–25 branch can, *if* the meter is quiet at zero.

11. Electrical Architecture

Water Guardian — wiring (TUF-2000M → RS485 → ESP32)

Verified: meter power 8–36 VDC @ 50 mA, isolated RS485 port, Modbus RTU 9600 8N1 (manual v13.44). Module pin names vary — check your board.



(VERIFIED interface facts; module choice ASSUMED best practice.)

- The meter's RS485 is a **differential bus** (A/B, ± 1.5 –5 V), galvanically isolated inside the meter; it is **not** a 12/24 V logic signal and must not be wired to the ESP32 directly.
- The meter is powered at **12 or 24 VDC** (8–36 V); the ESP32 gets 5 V from an LM2596 buck on the same rail. Never mains to the module.
- Interface options compared ([research/notes/alternatives_and_procurement.md](#) Task C): plain 5 V **MAX485** boards drive a 5 V RO output into the ESP32 RX — out of spec, avoid; **MAX3485** is native 3.3 V (DE/RE tied to one GPIO, $\approx$ SAR 25–45, on amazon.sa); **isolated auto-direction DFR0845** (ADM2483-class, 3 kV, $\approx$ SAR 150–200 landed) is the recommended primary for a mains-fed installation. 120 Ω termination at the far end only; short runs may omit it.
- Wiring diagram: [diagrams/wiring.svg](#) (power, grounds, A/B, RX/TX, DE/RE, isolation note).

12. ESP32 Integration

Firmware prototype in [firmware/esp32_prototype/](#) (PlatformIO, Arduino core), status **compiles; logic host-tested; not yet run on a real meter**:

- Modbus RTU master in-tree (FC03/06, CRC16, timeouts, exception/CRC/bad-frame results) with an injected transport so it runs on hardware and in host tests.
- `tuf2000m.*`: register map from the manual (1-based → PDU –1), REAL4 low-word-first decode with a runtime toggle and a **self-check on register 221 (inner diameter)**, validity policy (step = R, error bits 0–3 clear, $Q \geq 60$, strength $\geq \sim 60$ %, ratio 97–103 %, finite flow).
- `rule_engine.*`: C++ port of the simulator engine — same parameters, moving average, gap tolerance, NVS-persistable incident + 32-event unsynced buffer.
- `main.cpp`: 1 s polling, NTP-based schedule with a **conservative policy when the clock is invalid**, serial-console mode override for the demo, non-blocking Wi-Fi reconnection, Telegram + optional HTTP sync with ordered retry, periodic NVS checkpoint.
- Host tests `test/test_native`: **37/37 pass** (CRC vector, word order, Modbus OK/timeout/CRC/exception, "no-signal zero is not valid", quiet persistence, unknown-keeps-incident, burst, occupied REVIEW, cleaning expiry, invalid clock, persist/restore without duplicate alert, gap tolerance).

- `tools/virtual_tuf2000m.py` : a Modbus RTU **slave** answering with the meter's register layout and fault injection (no signal, poor Q, dropped/corrupted frames, word order) for end-to-end tests over a socat pty pair or a USB-RS485 adapter.

13. Modbus Protocol

VERIFIED from the manual (p.39–47) and three code bases:

Item	Value
Function codes	0x03 read holding, 0x06 write single — only these
Serial	9600 8N1 default (M62); address via keypad only (M46), default 1
Protocol default	MODBUS-ASCII — set M63 = 1 for RTU
Addressing	manual register N → PDU address N–1
Flow rate	reg 1–2, REAL4, m³/h (unit reg 1437)
Velocity	reg 5–6, REAL4 m/s
Totalizers	reg 9–10 LONG + 11–12 REAL4 fraction ($\times$ unit reg 1438 $\times 10^{(\text{reg } 1439-3)}$); simplest: reg 113–118 REAL4 in m³
Temperature	reg 33–34 / 35–36 REAL4 °C (PT100 fitted only)
Signal quality	reg 92 low byte 0–99; working step letter in high byte
Strength	reg 93 / 94, 0–2047 (display scale 0–99.9; mapping not published)
Transit ratio	reg 97–98 REAL4 %, normal 100 ± 3
Error bits	reg 72 (bit0 no signal, bit1 low, bit2 poor, bit3 empty pipe, bit4 hardware ... bit15 AI over-range)
Inner diameter	reg 221–222 REAL4 mm (used as word-order self-check)
Word order	low word first (CDAB) per AiltonFidelix, jkkorpi70, arivin29 (worked example $36.625\text{ °C} = 0x4212\ 8000 \rightarrow$ regs $[0x8000, 0x4212]$); one blog says "big-endian" → CONFLICT, handled by toggle + self-check
Variants	M/B/H/F/S/P share the table; register set may differ on clones (C3 note) → validate on the actual unit

14. Simulation Architecture

Tool survey (`research/notes/simulation_tools.md`): Wokwi can run the ESP32 firmware and a C custom chip can act as a Modbus slave; pymodbus/socat give a virtual meter on a serial port; PlatformIO native / ESP-IDF host target unit-test the logic; QEMU has UART but no Wi-Fi; Tinkercad/Proteus/SimulIDE cannot run ESP32 + Modbus; **no general-purpose tool simulates transit-time acoustics through a pipe wall** (COMSOL/k-Wave are research-grade and out of scope). We therefore split the system:

Layer	Simulated by	Fidelity
A Hydraulic / flow	<code>engine/hydraulics.js</code>	CALCULATED (geometry only; no transients)
B Sensor	<code>engine/sensorModel.js</code> — configurable behaviour model (cutoff, noise, error, zero drift, Q/strength, step letters, comm faults); profiles IDEAL / CONSERVATIVE / TUF-2000M published / TUF-2000M field-reports / CUSTOM	MODEL — no physics ; every number is an assumption except the published ones
C Modbus / comms	<code>tools/virtual_tuf2000m.py</code> + firmware <code>modbus_rtu</code> host tests	SIMULATED (protocol, framing, faults; not wire electrical)
D ESP32 firmware	logic host-tested; sketch compiles	SIMULATED (host) — not run on silicon yet
E Detection logic	<code>engine/ruleEngine.js</code> $\equiv$ <code>rule_engine.cpp</code>	SIMULATED, deterministic
F Dashboard / notification	simulator phone emulator; firmware Telegram optional	EMULATED

Interactive bench: `simulator/index.html` (open directly). Water moves at the true velocity; every control from the brief is present; rules are editable; the "Break it" panel injects the 15 failure modes.

15. Simulator Screenshots

`docs/screenshots/01_idle.png` (bench idle) · `02_incident.png` (3 L/min quiet → UNEXPECTED_FLOW, incident card, phone alert) · `03_sensor_unknown.png` (RS485 disconnected during the incident → SENSOR_UNKNOWN, incident preserved) · `04_offline.png` (Wi-Fi lost → OFFLINE overlay, events queued) · `05_hydraulics.png` (velocity table coloured against the profile's minimum) · `06_scenarios.png` (campaign run in-browser).

Water Guardian — Digital Test Bench

Tank → Pipe → Clamp-on ultrasonic sensor (MODEL) → ESP32 → Rule engine → Dashboard / Alert. Sensor physics are **not** simulated; the sensor block is a configurable behaviour model.

▶ Run

■ Pause

↺ Reset

10×

PLANT & INSTALLATION

True flow rate0.00 L/min

0

0.25

0.5

1

2

5

Pipe presetPPR 25 mm PN20 (ID = 16.6)

Material

Internal Ø (mm)

Wall (mm)

PPR

16.6

4.2

Velocity in pipe: 0.000 m/s · area 216.4 mm²

Sensor conditions

Sensor quality (installation/pipe)100%

Air / bubbles0%

Attachment

Coupling

Straight pipe

Good

Good

Good

Sensor profile (model)

TUF-2000M (published spec + assumptions)

PUBLISHED (manual v13.44, research/notes/tuf2000m.md): accuracy "better than 1 %", repeatability "better than 0.2 %", measurement period 500 ms, M41 low-flow cutoff default 0.03 m/s, M40 damping default 10 s. Velocity range ±0.01-12 m/s is a VENDOR claim (not in the manual). ASSUMED: noiseAbsMs 0.005, zeroOffsetMs 0.005, commFailProb 0.5 %.

Min velocity (m/s)

Low-flow cutoff (m/s)

0.03

0.03

Accuracy (%)

Noise (m/s, 1σ)

Zero offset (m/s)

1

0.005

0.005

Update interval (s)

Comm fail prob.

0.5

0.005

Facility & links

Facility mode

Clock / schedule

Quiet

Valid

Sensor link (RS485)

Wi-Fi

Backend

Connected

Online

Up

Rules (editable)

SYSTEM

Tank

Clamp-on transducers (MODEL)

Flow meter
no reply

ESP32
NORMAL

Wi-Fi/Backend
ONLINE

Rule engine
0 min > thr

Dashboard/Phone
synced

Mode: QUIET · sim 00:00:00

NORMAL

EXPECTED_FLOW

PERSISTENT_FLOW

UNEXPECTED_FLOW

REVIEW

SENSOR_UNKNOWN

RESOLVED

OFFLINE

SIM CLOCK

METER READING

SIGNAL / Q / STEP

ABOVE THRESHOLD FOR

00:00:00

—

—

0.0 min

Event log (ESP32 NVS buffer)

Hydraulic comparison

Scenario runner

CARETAKER PHONE (NOTIFICATION EMULATOR)

No notifications yet.

Messages are queued locally while Wi-Fi/backend are down and delivered on reconnection. Real Telegram delivery is optional in the firmware (see firmware/README.md); this page never sends anything.

Dashboard

LINK

ONLINE

UNSYNCED EVENTS

0

SUSPECT LITRES (OPEN)

0

TRUE TOTAL (HIDDEN REF)

0.0 L

Legend: · MODEL: configurable assumption · CALC: pure geometry · LOGIC: deterministic firmware logic. Nothing here proves the real ultrasonic measurement on a real pipe.

Water Guardian — Digital Test Bench

Tank → Pipe → Clamp-on ultrasonic sensor (MODEL) → ESP32 → Rule engine → Dashboard / Alert. Sensor physics are **not** simulated; the sensor block is a configurable behaviour model.

▶ Run

■ Pause

↺ Reset

600×

PLANT & INSTALLATION

True flow rate3.00 L/min

0

0.25

0.5

1

2

5

Pipe presetPPR 25 mm PN20 (ID = 16.6)

Material

Internal Ø (mm)

Wall (mm)

PPR

16.6

4.2

Velocity in pipe: 0.231 m/s · area 216.4 mm²

Sensor conditions

Sensor quality (installation/pipe)100%

Air / bubbles0%

Attachment

Coupling

Straight pipe

Good

Good

Good

Sensor profile (model)

TUF-2000M (published spec + assumptions)

PUBLISHED (manual v13.44, research/notes/tuf2000m.md): accuracy "better than 1 %", repeatability "better than 0.2 %", measurement period 500 ms, M41 low-flow cutoff default 0.03 m/s, M40 damping default 10 s. Velocity range ±0.01-12 m/s is a VENDOR claim (not in the manual). ASSUMED: noiseAbsMs 0.005, zeroOffsetMs 0.005, commFailProb 0.5 %.

Min velocity (m/s)

Low-flow cutoff (m/s)

0.03

0.03

Accuracy (%)

Noise (m/s, 1σ)

Zero offset (m/s)

1

0.005

0.005

Update interval (s)

Comm fail prob.

0.5

0.005

Facility & links

Facility mode

Clock / schedule

Quiet

Valid

Sensor link (RS485)

Wi-Fi

Backend

Connected

Online

Up

Rules (editable)

SYSTEM

Tank

Clamp-on transducers (MODEL)

Flow meter
3.17 L/min

ESP32
UNEXPECTED_FLOW

Wi-Fi/Backend
ONLINE

Rule engine
22 min > thr

Dashboard/Phone
synced

Mode: QUIET · sim 00:22:29

NORMAL

EXPECTED_FLOW

PERSISTENT_FLOW

UNEXPECTED_FLOW

REVIEW

SENSOR_UNKNOWN

RESOLVED

OFFLINE

SIM CLOCK

METER READING

SIGNAL / Q / STEP

ABOVE THRESHOLD FOR

00:22:29

3.17 L/min

72.2/71.9 · Q69 · R

22.5 min

Event log (ESP32 NVS buffer)

Hydraulic comparison

Scenario runner

Unexpected Continuous Water Use

Measured flow3.06 L/min (peak 3.11)

Duration22.5 min

Facility modeQUIET

Sensor quality69%

Estimated suspect consumption69.0 litres

Reason"Flow remained above 0.5 L/min for 10 minutes during a QUIET period."

StatusOpen — awaiting caretaker

ACKNOWLEDGE

EXPECTED ACTIVITY

INCIDENT CHECKED

This is a persistent-unexpected-flow claim, not a leak diagnosis. A single aggregate meter cannot identify the fixture.

00:10:01 #1 UNEXPECTED_FLOW ("14":1,"flowLpm":3.1061612903225793,"durationMin":10,"mode":"QUIET","quality":69,"reason":"Flow remained above 0.5 L/min for 10 minutes during a QUIET period.") ✓

CARETAKER PHONE (NOTIFICATION EMULATOR)

delivered 00:10:01

Unexpected Water Consumption

Location: Mosque Pilot

Monitored Zone: Ablution Supply

Flow: 3.1 L/min

Duration: 10 min

Mode: QUIET

Sensor: Marginal (Q=69)

Reason: Flow remained above 0.5 L/min for 10 minutes during a QUIET period.

Please inspect the facility.

Messages are queued locally while Wi-Fi/backend are down and delivered on reconnection. Real Telegram delivery is optional in the firmware (see firmware/README.md); this page never sends anything.

Dashboard

LINK

ONLINE

UNSYNCED EVENTS

0

SUSPECT LITRES (OPEN)

69.0 L

TRUE TOTAL (HIDDEN REF)

67.4 L

Legend: · MODEL: configurable assumption · CALC: pure geometry · LOGIC: deterministic firmware logic. Nothing here proves the real ultrasonic measurement on a real pipe.

Air / bubbles

0%

Attachment

Coupling

Straight pipe

Good

Good

Good

Sensor profile (model)

TUF-2000M (published spec + assumptions)

PUBLISHED (manual v13.44, research/notes/tuf2000m.md); accuracy "better than 1 %", repeatability "better than 0.2 %", measurement period 500 ms, M41 low-flow cutoff default 0.03 m/s, M40 damping default 10 s. Velocity range ±0.01–12 m/s is a VENDOR claim (not in the manual). ASSUMED: noiseAbsMs 0.005, zeroOffsetMs 0.005, commFailProb 0.5 %.

Min velocity (m/s)

Low-flow cutoff (m/s)

0.03

0.03

Accuracy (%)

Noise (m/s, 1σ)

Zero offset (m/s)

1

0.005

0.005

Update interval (s)

Comm fail prob.

0.5

0.005

Facility & links

Facility mode

Clock / schedule

Quiet

Valid

Sensor link (RS485)

Wi-Fi

Backend

Disconnect

Online

Up

Rules (editable)

Flow threshold (L/min)

Hysteresis (%)

0.5

20

Quiet persistence (min)

Occupied persistence (min)

10

45

Cleaning override (min)

Resolve after (min)

60

3

Quality threshold

Unknown grace (s)

60

15

Break it

Sensor disconnect

Kill Wi-Fi

Kill backend

Poor signal

Air 60%

Mount → Poor

ESP32 reboot

Invalid clock

Flow burst 8 L/min 30 s

Restore all

Mode: QUIET · sim 00:31:09

NORMAL

EXPECTED_FLOW

PERSISTENT_FLOW

UNEXPECTED_FLOW

REVIEW

SENSOR_UNKNOWN

SIM CLOCK

00:31:09

METER READING

NO COMM

SIGNAL / Q / STEP

—

ABOVE THRESHOLD FOR

31.1 min

Unexpected Continuous Water Use

Measured flow

3.05 L/min (peak 3.11)

Duration

24.5 min

Facility mode

QUIET

Sensor quality

69%

Estimated suspect consumption

75.1 litres

Reason

"Flow remained above 0.5 L/min for 10 minutes during a QUIET period."

Status

Open — awaiting caretaker

ACKNOWLEDGE

EXPECTED ACTIVITY

INCIDENT CHECKED

This is a persistent-unexpected-flow claim, not a leak diagnosis. A single aggregate meter cannot identify the fixture.

Event log (ESP32 NVS buffer)

Hydraulic comparison

Scenario runner

00:10:01 #1 UNEXPECTED_FLOW {"id":1,"flowLpm":3.1061612983225793,"durationMin":10,"mode":"QUIET","quality":69,"reason":"Flow remained above 0.5 L/min for 10 minutes during a QUIET period."} ✓

00:24:45 #2 SENSOR_UNKNOWN {"reason":"NO_RESPONSE (timeout)","incidentOpen":true} ✓

📡 delivered 00:24:45

🔴 Sensor status UNKNOWN (NO_RESPONSE (timeout)).

Water use cannot be confirmed as zero.

Incident open: yes.

Messages are queued locally while Wi-Fi/backend are down and delivered on reconnection. Real Telegram delivery is optional in the firmware (see firmware/README.md); this page never sends anything.

Dashboard

LINK

ONLINE

UNSYNCED EVENTS

0

SUSPECT LITRES (OPEN)

75.1 L

TRUE TOTAL (HIDDEN REF)

93.4 L

Legend: MODEL · configurable assumption · CALC · pure geometry · LOGIC · deterministic firmware logic. Nothing here proves the real ultrasonic measurement on a real pipe.

Air / bubbles

0%

Attachment

Coupling

Straight pipe

Good

Good

Good

Sensor profile (model)

TUF-2000M (published spec + assumptions)

PUBLISHED (manual v13.44, research/notes/tuf2000m.md); accuracy "better than 1 %", repeatability "better than 0.2 %", measurement period 500 ms, M41 low-flow cutoff default 0.03 m/s, M40 damping default 10 s. Velocity range ±0.01–12 m/s is a VENDOR claim (not in the manual). ASSUMED: noiseAbsMs 0.005, zeroOffsetMs 0.005, commFailProb 0.5 %.

Min velocity (m/s)

Low-flow cutoff (m/s)

0.03

0.03

Accuracy (%)

Noise (m/s, 1σ)

Zero offset (m/s)

1

0.005

0.005

Update interval (s)

Comm fail prob.

0.5

0.005

Facility & links

Facility mode

Clock / schedule

Quiet

Valid

Sensor link (RS485)

Wi-Fi

Backend

Connected

Offline

Up

Rules (editable)

Flow threshold (L/min)

Hysteresis (%)

0.5

20

Quiet persistence (min)

Occupied persistence (min)

10

45

Cleaning override (min)

Resolve after (min)

60

3

Quality threshold

Unknown grace (s)

60

15

Break it

Sensor disconnect

Kill Wi-Fi

Kill backend

Poor signal

Air 60%

Mount → Poor

ESP32 reboot

Invalid clock

Flow burst 8 L/min 30 s

Restore all

Mode: QUIET · sim 00:40:29

NORMAL

EXPECTED_FLOW

PERSISTENT_FLOW

UNEXPECTED_FLOW

REVIEW

SENSOR_UNKNOWN

SIM CLOCK

00:40:29

METER READING

3.24 L/min

SIGNAL / Q / STEP

70.6/69.5 · Q69 · R

ABOVE THRESHOLD FOR

40.5 min

Unexpected Continuous Water Use

Measured flow

3.05 L/min (peak 3.14)

Duration

40.5 min

Facility mode

QUIET

Sensor quality

69%

Estimated suspect consumption

97.0 litres

Reason

"Flow remained above 0.5 L/min for 10 minutes during a QUIET period."

Status

Open — awaiting caretaker

ACKNOWLEDGE

EXPECTED ACTIVITY

INCIDENT CHECKED

This is a persistent-unexpected-flow claim, not a leak diagnosis. A single aggregate meter cannot identify the fixture.

Event log (ESP32 NVS buffer)

Hydraulic comparison

Scenario runner

00:10:01 #1 UNEXPECTED_FLOW {"id":1,"flowLpm":3.1061612983225793,"durationMin":10,"mode":"QUIET","quality":69,"reason":"Flow remained above 0.5 L/min for 10 minutes during a QUIET period."} ✓

00:24:45 #2 SENSOR_UNKNOWN {"reason":"NO_RESPONSE (timeout)","incidentOpen":true} ✓

00:33:20 #3 SENSOR_RECOVERED {} ✓

00:33:50 #4 LINK {"Link":"OFFLINE"} 🚨 unsynced

📡 delivered 00:24:45

🔴 Sensor status UNKNOWN (NO_RESPONSE (timeout)).

Water use cannot be confirmed as zero.

Incident open: yes.

Messages are queued locally while Wi-Fi/backend are down and delivered on reconnection. Real Telegram delivery is optional in the firmware (see firmware/README.md); this page never sends anything.

Dashboard

LINK

OFFLINE

UNSYNCED EVENTS

1

SUSPECT LITRES (OPEN)

97.0 L

TRUE TOTAL (HIDDEN REF)

121.4 L

Legend: MODEL · configurable assumption · CALC · pure geometry · LOGIC · deterministic firmware logic. Nothing here proves the real ultrasonic measurement on a real pipe.

Water Guardian — Digital Test Bench

Tank → Pipe → Clamp-on ultrasonic sensor (MODEL) → ESP32 → Rule engine → Dashboard / Alert. Sensor physics are **not** simulated; the sensor block is a configurable behaviour model.

▶ Run

■ Pause

↺ Reset

600 ▾

PLANT & INSTALLATION

True flow rate3.00 L/min

0

0.25

0.5

1

2

5

Pipe preset

PPR 25 mm PN20 (ID = 16.6)

Material

PPR

Internal Ø (mm)

16.6

Wall (mm)

4.2

Velocity in pipe: 0.231 m/s · area 216.4 mm²

Sensor conditions

Sensor quality (installation/pipe)100%

Air / bubbles0%

Attachment

Good

Coupling

Good

Straight pipe

Good

Sensor profile (model)

TUF-2000M (published spec + assumptions)

PUBLISHED (manual v13.44, research/notes/tuf2000m.md): accuracy "better than 1 %", repeatability "better than 0.2 %", measurement period 500 ms, M41 low-flow cutoff default 0.03 m/s, M40 damping default 10 s. Velocity range ±0.01-12 m/s is a VENDOR claim (not in the manual). ASSUMED: noiseAbsMs 0.005, zeroOffsetMs 0.005, commFailProb 0.5 %.

Min velocity (m/s)

0.03

Low-flow cutoff (m/s)

0.03

Accuracy (%)

1

Noise (m/s, 1σ)

0.005

Zero offset (m/s)

0.005

Update interval (s)

0.5

Comm fail prob.

0.005

Facility & links

Facility mode

Quiet

Clock / schedule

Valid

Sensor link (RS485)

Connected

Wi-Fi

Offline

Backend

Up

Rules (editable)

SYSTEM

Tank

Clamp-on transducers (MODEL)

Flow meter2.84 L/min

ESP32UNEXPECTED_FLOW

Wi-Fi/BackendOFFLINE

Rule engine45 min > thr

Dashboard/Phone1 queued

Mode: QUIET · sim 00:44:59

NORMAL

EXPECTED_FLOW

PERSISTENT_FLOW

UNEXPECTED_FLOW

REVIEW

SENSOR_UNKNOWN

RESOLVED

OFFLINE

SIM CLOCK00:44:59

METER READING2.84 L/min

SIGNAL / Q / STEP68.5/66.3 · Q66 · R

ABOVE THRESHOLD FOR45.0 min

Unexpected Continuous Water Use

Measured flow3.07 L/min (peak 3.14)

Duration45.0 min

Facility modeQUIET

Sensor quality69%

Estimated suspect consumption110.7 litres

Reason"Flow remained above 0.5 L/min for 10 minutes during a QUIET period."

StatusOpen — awaiting caretaker

ACKNOWLEDGE

EXPECTED ACTIVITY

INCIDENT CHECKED

This is a persistent-unexpected-flow claim, not a leak diagnosis. A single aggregate meter cannot identify the fixture.

Event log (ESP32 NVS buffer)

Hydraulic comparison

Scenario runner

$Q = v \cdot A$, $A = \pi D^2/4$. Cells colour = true velocity vs the selected profile's min velocity (green ≥ 2 × min - yellow ≥ min - red < min). This is geometry, not a measurement.

L/min ↓ / ID mm →	13.2	16.6 *	21.2	26.6	33.2	52.5	
0.25		0.0304 m/s	0.0193 m/s	0.0118 m/s	0.0075 m/s	0.0048 m/s	0.0019 m/s
0.5		0.0609 m/s	0.0385 m/s	0.0236 m/s	0.0150 m/s	0.0096 m/s	0.0038 m/s

CARETAKER PHONE (NOTIFICATION EMULATOR)

delivered 00:10:01

Unexpected Water Consumption

Location: Mosque Pilot

Monitored Zone: Ablution Supply

Flow: 3.1 L/min

Duration: 10 min

Mode: QUIET

Sensor: Marginal (Q=69)

Reason:
Flow remained above 0.5 L/min for 10 minutes during a QUIET period.

Please inspect the facility.

delivered 00:24:45

Sensor status UNKNOWN (NO_RESPONSE (timeout)).

Water use cannot be confirmed as zero.

Incident open: yes.

Dashboard

LINKOFFLINE

UNSYNCED EVENTS1

SUSPECT LITRES (OPEN)110.7 L

TRUE TOTAL (HIDDEN REF)134.9 L

Legend: · MODEL: configurable assumption · CALC: pure geometry · LOGIC: deterministic firmware logic. Nothing here proves the real ultrasonic measurement on a real pipe.

Water Guardian — Digital Test Bench

Tank → Pipe → Clamp-on ultrasonic sensor (MODEL) → ESP32 → Rule engine → Dashboard / Alert. Sensor physics are **not** simulated; the sensor block is a configurable behaviour model.

▶ Run

■ Pause

↺ Reset

600 ▾

PLANT & INSTALLATION

True flow rate3.00 L/min

0

0.25

0.5

1

2

5

Pipe preset

PPR 25 mm PN20 (ID = 16.6)

Material

PPR

Internal Ø (mm)

16.6

Wall (mm)

4.2

Velocity in pipe: 0.231 m/s · area 216.4 mm²

Sensor conditions

Sensor quality (installation/pipe)100%

Air / bubbles0%

Attachment

Good

Coupling

Good

Straight pipe

Good

Sensor profile (model)

TUF-2000M (published spec + assumptions)

PUBLISHED (manual v13.44, research/notes/tuf2000m.md): accuracy "better than 1 %", repeatability "better than 0.2 %", measurement period 500 ms, M41 low-flow cutoff default 0.03 m/s, M40 damping default 10 s. Velocity range ±0.01-12 m/s is a VENDOR claim (not in the manual). ASSUMED: noiseAbsMs 0.005, zeroOffsetMs 0.005, commFailProb 0.5 %.

Min velocity (m/s)

0.03

Low-flow cutoff (m/s)

0.03

Accuracy (%)

1

Noise (m/s, 1σ)

0.005

Zero offset (m/s)

0.005

Update interval (s)

0.5

Comm fail prob.

0.005

Facility & links

Facility mode

Quiet

Clock / schedule

Valid

Sensor link (RS485)

Connected

Wi-Fi

Offline

Backend

Up

Rules (editable)

SYSTEM

Tank

Clamp-on transducers (MODEL)

Flow meter3.07 L/min

ESP32UNEXPECTED_FLOW

Wi-Fi/BackendOFFLINE

Rule engine88 min > thr

Dashboard/Phone1 queued

Mode: QUIET · sim 01:27:59

NORMAL

EXPECTED_FLOW

PERSISTENT_FLOW

UNEXPECTED_FLOW

REVIEW

SENSOR_UNKNOWN

RESOLVED

OFFLINE

SIM CLOCK01:27:59

METER READING3.07 L/min

SIGNAL / Q / STEP67.7/66.4 · Q71 · R

ABOVE THRESHOLD FOR88.0 min

Unexpected Continuous Water Use

Measured flow3.06 L/min (peak 3.14)

Duration88.0 min

Facility modeQUIET

Sensor quality69%

Estimated suspect consumption242.0 litres

Reason"Flow remained above 0.5 L/min for 10 minutes during a QUIET period."

StatusOpen — awaiting caretaker

ACKNOWLEDGE

EXPECTED ACTIVITY

INCIDENT CHECKED

This is a persistent-unexpected-flow claim, not a leak diagnosis. A single aggregate meter cannot identify the fixture.

Event log (ESP32 NVS buffer)

Hydraulic comparison

Scenario runner

Run all 15 scenarios (headless, fast)

24/25 PASS (profile: TUF-2000M (published spec + assumptions))

#	Scenario	Expected	Result	Detail
1	No flow during quiet period	No alert	PASS	states=NORMAL
2	Short legitimate 5 L/min burst (90 s) in quiet	No persistent-flow alert	PASS	states=NORMAL,PERSISTENT_FLOW, maxAbove=2.4 min

CARETAKER PHONE (NOTIFICATION EMULATOR)

delivered 00:10:01

Unexpected Water Consumption

Location: Mosque Pilot

Monitored Zone: Ablution Supply

Flow: 3.1 L/min

Duration: 10 min

Mode: QUIET

Sensor: Marginal (Q=69)

Reason:
Flow remained above 0.5 L/min for 10 minutes during a QUIET period.

Please inspect the facility.

delivered 00:24:45

Sensor status UNKNOWN (NO_RESPONSE (timeout)).

Water use cannot be confirmed as zero.

Incident open: yes.

Dashboard

LINKOFFLINE

UNSYNCED EVENTS1

SUSPECT LITRES (OPEN)110.7 L

TRUE TOTAL (HIDDEN REF)134.9 L

Legend: · MODEL: configurable assumption · CALC: pure geometry · LOGIC: deterministic firmware logic. Nothing here proves the real ultrasonic measurement on a real pipe.

16. Virtual Test Results

tests/run.js → tests/results/CAMPAIGN_RESULTS.md (deterministic seed, 1 s step). **SIMULATED.**

Run	Profile	Result	Reading
TUF2000M_PUBLISHED	published spec + assumed noise	24/25	only rung 3 (0.25 L/min on ID 16.6 = 0.019 m/s) fails — below the 0.03 m/s cutoff by geometry
CONSERVATIVE	0.05 m/s cutoff, 3 %	24/25	same
IDEAL	no cutoff	24/25	"fails" 13 only because an ideal sensor <i>does</i> see 0.5 L/min on DN50 — a finding, not a defect
FIELD_REPORTS_RAW (stress)	+0.03 m/s zero offset, 0.05 m/s noise, no zero-cal	18/25	meter reads 0.33 L/min at true zero ; false incidents at 0.15/0.30 L/min thresholds; incidents at 0.5 L/min never resolve (§17)
FIELD_REPORTS_ZEROCAL (stress)	same noise, zero-cal applied	23/25	no false incidents; resolves normally; transient PERSISTENT_FLOW blips at zero; single samples can read 0 while flowing (averaging covers it)
FIELD_REPORTS_THR1 (stress)	no zero-cal, threshold 1.0 L/min	20/25	incidents resolve; no false incident at 1.0 (the two "failures" are ladder rungs that force their own low thresholds)

Logic failures across all runs: **0**. Firmware host tests: 37/37.

Scenario coverage vs the brief (1–15 all present, plus 9b cleaning expiry, 16 invalid clock, 17 burst, 18 air, 19 mount degrades, 20 Modbus CRC errors, 21 zero-drift stress, and zero-flow controls 3z/4z/5z paired with the ladder rungs).

17. Failure-Mode Tests

#	Failure	Behaviour proven (SIMULATED)
1	Sensor disconnected	NO_RESPONSE → SENSOR_UNKNOWN after 15 s grace; open incident frozen, not resolved; recovers with incident intact
2	RS485 CRC/timeouts (5 %)	absorbed by grace + averaging; detection unaffected
3	Wi-Fi lost	detection at 600 s while offline; events queued; synced on reconnect (0 unsynced at end)
4	Backend down	same as Wi-Fi (link overlay OFFLINE)
5	Poor signal (Q < 60)	SENSOR_UNKNOWN, no numeric claim
6	Air 80 %	step K/H → SENSOR_UNKNOWN; never a valid zero while flowing

#	Failure	Behaviour proven (SIMULATED)
7	Mount → Poor mid-incident	degrades to SENSOR_UNKNOWN; incident not silently closed
8	Flow below detection	0.25 L/min on DN25 undetected — reported as BELOW_PROFILE_USABLE_RANGE
9	Large pipe + small flow	DN50 at 0.5 L/min = 0.004 m/s → correctly no claim; blind spot documented
10	Cleaning in quiet hours	EXPECTED_FLOW; override expiry logged; after expiry the QUIET rule applies
11	Long legitimate occupied flow	EXPECTED_FLOW → REVIEW (low priority) only after 45 min; never a quiet alert
12	ESP32 reboot	incident restored from NVS; one alert total; resolves later
13	Invalid clock	conservative: treated as occupied → no quiet alert
14	Sudden burst	PERSISTENT_FLOW blip, no incident
15	Persistent moderate flow	UNEXPECTED_FLOW after 10 min; RESOLVED 3 min after flow stops; litres estimated

Stress finding (the important one): **an uncalibrated noisy meter makes incidents that cannot resolve** because the "zero" sits at 0.33 L/min against a 0.40 L/min resolve level. Mitigations shown to work in the model: M42 zero calibration saved with M26, 30 s averaging, and/or threshold ≥ 1.0 L/min on DN25. These become Phase 7 acceptance criteria.

18. Physical Validation Plan

Water Guardian — physical validation rig (Phase 7)

Gravity or mains-fed loop. Reference = timed gravimetric collection (scale, 1 g) — independent of the meter under test.

Header tank 20 L

Test pipe: PPR 25 mm PN20 (ID ≈ 16.6) and 32 mm ID — also try 1" GI for comparison

TS-2 pair, V-method, on the SIDE of the pipe

air bleed (keep pipe FULL)

isolation valve

$\geq 10 D$ upstream straight (170-210 mm)

$\geq 5 D$ downstream

flow-setting valve (needle/ball, sets 0.25-5 L/min)

TUF-2000M + ESP32
log flow, Q, strength, ratio

Rules of the rig

1. Pipe completely full; outlet above the transducer level so the section never drains.
2. Zero calibration (M42) with BOTH valves shut and water still; solidify with M26.
3. Each rung: 3 timed collections (≥ 60 s or ≥ 1 L); each mounting: remove, re-couple, re-clamp — 5 mountings.
4. Record Q, strength up/down, M91 ratio, working step at every reading. A reading with $Q < 60$ is logged as INVALID, not as zero.
5. Disturbance trials: tap the pipe, open/close a downstream tap briefly, drain to half-full — count false incidents.

Pass targets (ours, not a standard): $\geq 19/20$ detections at the declared threshold; ≤ 1 false incident in 40 mixed trials.

PHYSICAL_TEST_PLAN.md — preconditions (M63 RTU, measured pipe parameters, V-method, full pipe, M42+M26 zero), flow ladder 0/0.25/0.5/1/2/5 L/min with three timed gravimetric collections per rung and **five independent mountings**, a 30 min zero-stability log, a 20-trial detection run at the declared threshold (target $\geq 19/20$), a 40-trial mixed false-incident run (target $\leq 1/40$), pre-declared abandon criteria, and a CSV data sheet (tests/results/physical_test_template.csv). Rig drawing: diagrams/hydraulic_layout.svg .

19. Complete BOM

BOM.md . Essentials: meter + TS-2 (SAR $\approx$ 430–1,090 landed), RS485 module (25–200), 12/24 V PSU (19–120), LM2596 (27), couplant (40–98), clamps (45), wiring (20–30), enclosure (35–96); rig: PPR 25/32 mm + fittings ($\approx$ 60–90), valves (50–90), 20 L can (85), jug (8–15), 1 g scale (19–44).

20. Saudi Price Estimates

Scenario	SAR
Rig + electronics, meter borrowed	$\approx$ 450–700
Rig + AliExpress TUF-2000M-TS-2 (standard shipping)	$\approx$ 900–1,300
+ DHL + isolated DFR0845	$\approx$ 1,300–2,000
Ceiling (2,300)	not exceeded in any configuration

Prices are snippet-level (VERIFIED listings, ESTIMATE totals); 15 % VAT applies to imports; duty waived < SAR 1,000.

21. Procurement Recommendations

1. **Ask for a loan/demo first, in parallel with obtaining the pipe facts:** KFUPM ME lab (HM 500 bench for calibration), IAU fluid lab, KSU hydraulics; distributors E+H Arabia, KROHNE KSA, IICS–Abunayyan, Kanoo Energy, Flexim ME. Give them a one-page pilot framing.
2. If no loan in $\sim$ 2 weeks: order **TUF-2000M + TS-2 with RS485** from AliExpress (item 32518776067-class listing, confirm "TS-2 DN15–100" and "RS485" in the seller chat), standard shipping, $\approx$ SAR 450–600 landed. Order a **KNACRO MAX3485** locally now (cheap, arrives in days) and decide on DFR0845 after the bench test.
3. Buy the rig items locally now (they are useful regardless of meter).
4. Do **not** buy the handheld TUF-2000H as the product sensor (no Modbus); it is acceptable only as a survey instrument if it can be returned within 15 days.
5. Budget the meter as a **test article**: a negative result is a legitimate, reportable outcome of the pre-purchase programme.

22. Risks

Risk	Likelihood	Impact	Mitigation
TS-2 cannot achieve $Q \geq 60$ on PPR PN20	medium-high (no PPR evidence; soft, thick wall)	fatal for PPR sites	test on PPR <i>and</i> GI/PVC; W-method; sanding; alternative site pipe (metal riser branch)
Zero noise $\geq$ threshold on DN25	medium (reported on Cu 22–28 mm)	false alarms / unresolvable incidents	M42+M26, 30 s average, threshold $\geq$ 1 L/min, declare the threshold honestly
Meter freezes last value (M28) or	certain unless configured	silent misreads	validity policy on step/Q/bits; set M28 = NO

Risk	Likelihood	Impact	Mitigation
reports 0 on signal loss			
Modbus ASCII default	certain	"no comms" on day 1	set M63 = 1
Word-order mismatch	low	garbage floats	reg-221 self-check + toggle
Wrong wall thickness entered	high	confident but wrong absolute flow	measure with calliper/ultrasonic gauge; bucket test; M45
Couplant dries	high over weeks	Q decay → SENSOR_UNKNOWN	grease-type couplant; monitor Q trend
Shipping delay / DOA / no return	medium	schedule	order early; Alibaba Trade Assurance; local MAX3485
Competition rules unverified	medium	wrong deliverable	open official pages now

23. Unknowns

Must obtain before buying: pipe **material** (PPR? PVC? GI?), **OD and wall/PN class**, ID, an accessible **full** horizontal straight run ($\geq 10D + 5D$) on the **branch** feeding the ablution area, the riser diameter, water temperature range, whether a caretaker can operate a Telegram bot, and the mosque's daily schedule (for the QUIET windows). Still unverified after this study: TS-2 minimum pipe (DN15 vs 25), PP-R sound speed for M15, strength scaling 0-2047 → 0-99.9, register set on clone firmware, competition deadline/criteria.

24. GO / PIVOT / NO-GO Decision

GO WITH CONDITIONS.

1. Feasible? **Yes for the system; unproven for the sensor at < 1 L/min on PPR.**
2. Proven: RS485/Modbus interface facts, register map, power, diagnostics; hydraulic implications; that the ESP32 chain and rules behave correctly under every failure mode we could model.
3. Simulated: layers C-F and the sensor *behaviour* envelope.
4. Physically unverified: acoustic coupling on PPR, zero stability, minimum detectable flow, couplant life.
5. Biggest risk: **zero-flow noise/drift on a small plastic pipe** (false or unresolvable incidents), closely followed by PPR coupling.
6. Minimum useful flow to test: **1 L/min** is the realistic declared threshold on DN25 PPR; test 0.5 and 0.25 to know the floor, but do not promise them.
7. Best location: the **DN20-32 branch to the ablution area**, horizontal, full, $\geq 10D$ straight upstream, accessible, away from the pump.
8. Pipe information: \$23.
9. Sensor sufficient? **Probably for $\geq 1-2$ L/min continuous flows** (running tap, failed float valve) — which is the product's actual value; **probably not for drips.**
10. Buy / borrow / other / pivot: **borrow first; buy the AliExpress kit if no loan within two weeks; keep Keyence FD-Q as the funded upgrade path if PPR coupling fails and budget can grow.**
11. What could make it fail completely: Q never ≥ 60 on the pilot's PPR pipe; zero noise > 2 L/min after calibration; no accessible full straight run.

12. Abandon this sensing approach if, after 5 mountings with correct setup: zero noise floor > 2 L/min on DN25, < 15/20 detections at 2 L/min, > 4/40 false incidents with mitigations, or Q < 60 unattainable on PPR (pre-declared in the test plan §5).

25. Exact Next Steps

1. **Today:** open the MEWA UIC page and SEU notice (unblocked browser); record deadline, platform, criteria. Send loan requests (template: pilot framing) to KFUPM ME lab, IAU, E+H Arabia, KROHNE KSA, IICS, Kanoo Energy.
2. **This week:** visit the pilot mosque; photograph and measure the branch and riser (calliper OD, PN class printed on the pipe); note straight lengths and access; get the prayer/cleaning schedule.
3. **This week:** buy locally: MAX3485 module, LM2596, 12 V adapter, PPR 25 + 32 mm bars and fittings, 2 valves, jerry can, jug, 1 g scale, couplant/grease. Build the rig (`hydraulic_layout.svg`).
4. **This week:** flash the firmware on the owned ESP32; run it against `tools/virtual_tuf2000m.py` over a USB-RS485 adapter (or a second ESP32/USB-TTL) to prove the RS485 path end-to-end before the meter arrives.
5. **Day 10-14:** if no loan is confirmed, order the TUF-2000M-TS-2 RS485 kit ($\approx$ SAR 450-600 landed).
6. **On meter arrival:** meter setup checklist (M63, M46, M11-16, M23-25, M41 default, M28 = NO, M42 → M26), then the Phase 7 ladder on PPR 25 → decide the declared threshold from data → 20/40 trials → write results into `tests/results/` .
7. **Then:** freeze the MVP claim ("persistent unexpected flow $\geq$ X L/min, verified on pipe Y") and build the poster/demo around the simulator + the bench rig.

26. Sources and URLs

SOURCES.md (44 entries, tagged by access level) and per-fact tags in `research/notes/*.md` .

Final question — "if the SAR 600+ were your own money, would you buy this sensor today?"

NO — BORROW FIRST. And if borrowing fails within two weeks: **YES — BUT ONLY AFTER** (a) the pilot pipe's material/OD/wall and a full, straight, accessible branch run are confirmed, (b) the order is the cheapest RS485-equipped TS-2 kit ($\approx$ SAR 450-600 landed, not SAR 1,000+), and (c) the team accepts in writing that a negative bench result is a valid deliverable.

Evidence: the interface, protocol and firmware side is verified to the register and pin level, so the *only* purchase risk is acoustic performance on a small PPR pipe at low velocity. On that risk the manual is silent about PPR, the hydraulic arithmetic puts 0.5 L/min right at the meter's default cutoff on DN25, and the only independent small-pipe reports describe zero-flow oscillation and drift — which our stress simulation shows would produce false and unresolvable incidents unless the zero is calibrated and the threshold is $\geq \sim 1$ L/min. A borrowed reference-class meter would answer that question for free; failing that, the cheap kit is worth buying *as an experiment* at the low landed price, provided nobody arrives at the competition claiming drip detection before the ladder test has been run.